Model-Based Dynamical Decoupling for Idle Superconducting Qubits

A Reproducible Synthetic Benchmark with Hardware Timing Constraints

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Christian De Zabala F.
Department of Computer Science
Department of Economics
Columbia University
cd3441@columbia.edu

Madeline Feldner
Department of Computer Science
Department of Statistics
Columbia University
mef2238@columbia.edu

Zarar Haider
Department of Computer Science
Columbia University
zh2555@columbia.edu

Udaya Vijay Anand
Department of Electrical Engineering
Columbia University
uv2125@columbia.edu

***Abstract*—An idle superconducting qubit waiting during a neighbour's mid-circuit measurement loses coherence to stochastic dephasing and to deterministic frequency shifts. We ask whether model-based optimization of dynamical-decoupling (DD) pulse timings and phases with CMA-ES outperforms strong canonical sequences. To answer it fairly, we pre-registered a synthetic transmon-like model, seeds, baselines and a training-only baseline-selection rule, and scored every arm on independent held-out noise. A finite-pulse Monte-Carlo simulator agrees with filter-function theory in five predetermined configurations (largest $|z| = 0.95$). Synthetic Ramsey/Hahn/CPMG spectroscopy recovers a predictive noise model (held-out RMSE $1.15\times$ the shot-noise floor), but the spectrum is not uniquely identified. The pre-registered Scenario A places canonical DD near the model's relaxation and white-noise floor, leaving limited optimization headroom; no CMA-ES mode delivered a practically meaningful gain over the selected XY4/XY8 baseline, whose infidelity (1.00×10^{-2} at $n = 16$) is already at the T_1-plus-white-noise floor (9.90×10^{-3}). With a deterministic Stark-like shift and virtual-Z correction, CMA-ES beat XY4 by a few percent at the nominal profile only at $n = 4$. The timing+phase nominal improvement did not consistently survive the pre-registered ±100 ns readout-timing shifts, and only phase-only sequences kept a marginal, model-specific advantage. Exploratory colored-noise ablations, with greater theoretical headroom, also showed no substantial held-out gain, and training gains did not generalize. All results are synthetic; no quantum hardware was used.**

***Index Terms*—dynamical decoupling, superconducting qubits, noise spectroscopy, CMA-ES, synthetic simulation, quantum control**

I. Introduction and project bounds

a) Motivation: An idle spectator qubit must preserve coherence while other qubits are measured and classical feedback is processed. Dynamical decoupling can suppress suitable low-frequency dephasing, but finite pulse errors, energy relaxation, and broadband fluctuations may limit improvement. We investigate whether model-specific pulse optimization provides gains beyond established XY4/XY8 sequences when the available error budget is already near its physical floor. The present analysis is a synthetic computational study; superconducting-hardware validation is a separate planned experiment.

b) Problem: An idle qubit accumulates phase errors from stochastic dephasing and, in dynamic circuits, from deterministic frequency shifts during neighbouring readout. We ask whether model-based optimization of DD sequences can reduce idle-window infidelity relative to strong canonical baselines under a fixed pulse budget and identical constraints.

c) Research questions:

- RQ1 Agreement of Monte Carlo coherence with analytic filter-function predictions under matched PSD conventions.
- RQ2 Predictive (held-out) accuracy of noise parameters fitted from synthetic Ramsey, Hahn and CPMG data.
- RQ3 Whether joint optimization of pulse timings and phases beats strong DD baselines in Scenario A (stochastic dephasing) on held-out noise.
- RQ4 Whether any gain over calibrated virtual-Z and standard DD survives uncertainty in readout-induced detuning (Scenario B).
- RQ5 Robustness to pulse imperfections and model mismatch. Additional drift, Stark rise-time and control-quantization tests remain future work.

d) Scope: Synthetic simulations only. Scenarios, seeds, baselines, the baseline-selection rule and the metrics were pre-registered in the repository before any optimization run; deviations are logged. A zero gain or a regression counts as a valid outcome. Metric: average gate infidelity against the corrected target (Sec. III).

II. Related work

Dynamical decoupling of open quantum systems was introduced by Viola et al. [1]. Uhrig [2] proposed optimized π-pulse sequences for keeping a qubit coherent, and Cywiński et al. [3] analysed how to enhance dephasing times in superconducting

qubits. Biercuk et al. [4] studied optimized DD in a model quantum memory. Biercuk, Doherty, and Uys described DD optimization as a filter-design problem [5], while Green and coauthors developed higher-order filter methods for driven gates [6]. These approaches directly motivate deterministic model-based objectives, which differ from the finite-trajectory CMA-ES objective used here. DD has also been used for noise spectroscopy: Bylander et al. [7] with a superconducting flux qubit and Álvarez and Suter [8] for colored noise. Robust DD was studied by Souza et al. [9], and Ezzell et al. [10] surveyed DD performance on superconducting qubits. Empirical or hardware-tailored learning approaches include Rahman et al. [11], Tong et al. [12] and Tong et al. [13].

a) Empirical versus model-based: Rahman et al. [11], Tong et al. [12], and Tong et al. [13] study empirical or hardware-tailored dynamical decoupling. In particular, Rahman et al. [11] optimizes rotational gates for superconducting hardware, while Tong et al. [13] studies empirical suppression in dynamic circuits. The present approach instead uses a fitted or assumed synthetic noise model as the optimization objective. Such a model-based approach reduces direct hardware calls but depends on model fidelity.

b) Positioning: This project is a model-based, synthetic-only study: sequences are optimized and evaluated against a simulated noise model, and no quantum hardware has been used. Its contribution is limited to a pre-registered, held-out synthetic benchmark of model-based DD optimization with explicit noise floors. Its main finding is negative: under the pre-registered synthetic model, CMA-ES did not provide a practically meaningful generalizable improvement over XY4/XY8 on held-out noise (Sec. V). A synthetic benchmark of this kind does not test whether its conclusions transfer to hardware.

III. Physical model and conventions

We use $\hbar = 1$, time in s, angular frequency and detuning in rad/s and angles in rad. In the rotating frame of the drive,

$$H(t) = \tfrac{1}{2}\big[\delta(t) + d(t) + \delta_{\text{off}}\big]\sigma_z + \tfrac{1}{2}\Omega(t)\big[\cos\varphi(t)\,\sigma_x + \sin\varphi(t)\,\sigma_y\big], \quad (1)$$

where $\delta(t)$ is zero-mean stationary noise, assumed *Gaussian* unless stated otherwise, $d(t)$ a known deterministic detuning profile (e.g. during neighbouring readout), and δ_{off} a static calibration offset.

a) Spectrum and filter: With autocorrelation $C(\tau) = \langle\delta(t+\tau)\delta(t)\rangle$ we use the *two-sided* PSD $S_2(\omega) = \int_{-\infty}^{\infty} C(\tau)e^{-i\omega\tau}\,d\tau$ in rad^2/s. For the switching function $y(t)$ (toggling-frame z component, ±1 between ideal π pulses),

$$Y(\omega) = \int_0^T y(t)e^{i\omega t}\,dt, \quad (2)$$

$$\chi = \operatorname{Var}\phi = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_2(\omega)\,|Y(\omega)|^2\,d\omega, \quad (3)$$

and, under the Gaussian assumption, $W = e^{-\chi/2}$ with pure-dephasing average gate infidelity $r = (1-W)/3$. Low- and high-frequency cutoffs of S_2 are stated explicitly for every model.

b) Energy relaxation: T_1 is modelled as amplitude damping with $\gamma = 1 - e^{-t/T_1}$ per free segment; channels are composed as Pauli transfer matrices, never by adding infidelities. Amplitude damping alone has average infidelity $r_{\text{AD}} = (2+\gamma-2\sqrt{1-\gamma})/6 \approx \gamma/3$ (derived in `docs/PHYSICS.md` and checked numerically against Haar and six-state averages).

c) Target and frame correction: Fidelity is evaluated against $U_{\text{target}} = R_z(\theta_{\text{vz}})U_0$, where U_0 is the ideal net control unitary of the sequence and θ_{vz} is a virtual-Z correction of the known $d(t)$ (zero unless frame correction is allowed). Average gate fidelity of a channel with PTM R is $F_{\text{avg}} = (\operatorname{Tr} R + 2)/6$ evaluated on $R_{\text{target}}^{\mathsf{T}}R$. For finite pulses we use the generalized (three-component toggling-frame) filter function, so $W = e^{-\chi/2}$ is exact only for instantaneous pulses and leading-order otherwise; the Monte-Carlo simulator needs neither approximation. Finite pulse widths, minimum spacing and timing-grid constraints are enforced for all sequences.

IV. Methods

a) Synthetic device (Scenario A): All device numbers are *synthetic assumptions*, fixed in a pre-registered plan (`docs/EXPERIMENT_PLAN.md`) before any optimization run. The idle window is $T = 2\,\mu$s (alternate $4\,\mu$s), with square π pulses of width $w = 32$ ns, a 4 ns guard and pulse starts on a 4 ns grid (an assumed controller tick). The dephasing PSD is the sum of a band-limited $1/f$ term ($2\pi\cdot 1$ Hz–$2\pi\cdot 10$ MHz, with $S_2(\omega) = A/|\omega|$, $A = 5.929\times10^9$, calibrated so that its isolated Ramsey coherence is e^{-1} at $10\,\mu$s), a Gaussian Ornstein–Uhlenbeck (Lorentzian) term with $\sigma = 1.236\times10^5$ rad/s and $\tau_c = 0.5\,\mu$s, and a white floor $S_0 = 10^4$ rad^2/s; $T_1 = 100\,\mu$s; and each trajectory draws a relative Rabi-amplitude error $\epsilon \sim \mathcal{N}(0, 0.02^2)$. Scenario B adds a deterministic Stark-like shift of $2\pi\cdot0.2$ MHz between 0.6 and 1.6 μs (20 ns tanh edges) as a stand-in for a neighbouring readout; it models a known frequency shift only, not measurement-induced dephasing.

b) Finite-pulse Monte-Carlo simulator: Gaussian noise trajectories are sampled on a 1 ns grid using hybrid spectral synthesis: a long periodic FFT for higher frequencies and explicit low-frequency random-phase quadrature below the inverse trajectory window (exact Ornstein–Uhlenbeck recursion for the Lorentzian; exact Markov jumps for telegraph noise). Free segments are exact z rotations by the integrated detuning; pulses are propagated in 8 piecewise-constant sub-steps, including detuning during the pulse. Amplitude damping is composed per free segment (exact, as it is phase-covariant) and approximated as $\mathrm{AD}(w/2)\,U\,\mathrm{AD}(w/2)$ during pulses. The noise-averaged Pauli transfer matrix gives F_{avg} against the corrected target. The reported ϵ is the mean of per-trajectory infidelities, which equals the channel infidelity because F_{avg} is linear in the channel. Unit tests check unitarity ($< 10^{-14}$), zero-noise compensated infidelity ($\le 10^{-10}$ for all families, odd and even n), $O(\Delta t^2)$ and $O(m^{-2})$ convergence in noise step and pulse sub-steps, and the T_1-only formula to 10^{-12}.

c) Spectroscopy and held-out fits: Synthetic Ramsey, Hahn, CPMG-2 and CPMG-4 data (14 log-spaced idle times

in $[0.1, 40]\,\mu$s, 1000 shots per point, binomial shot noise, an affine SPAM map) are fitted by binomial maximum likelihood with a parametric bootstrap. Two Gaussian models are fitted: *reduced* ($1/f$ with $\alpha = 1$, plus white) and *full* (reduced plus Lorentzian). T_1 is treated as known from a separate experiment, because white dephasing and $1/T_1$ are degenerate for instantaneous pulses. Held-out data are CPMG-8 and CPMG-16 at all times, and the training families at 13 unseen times, drawn with an independent generator. The fit never sees held-out observations; a unit test asserts identical parameters with and without test data. A misspecification control replaces the Lorentzian with telegraph noise of equal second-order statistics.

d) CMA-ES optimization: We use pycma 4.5.0 [14], an evolution strategy rather than reinforcement learning. Timings are encoded by n unconstrained coordinates, giving gaps $g_k = g_{\min} + (T - nw - (n+1)g_{\min})\,\mathrm{softmax}([0, z])_k$ with no redundant shift; starts are snapped to the grid, and candidates that cannot be repaired are rejected. Pulse phases are offsets from the base phases, projected so that the ideal net unitary stays equal to the base sequence's ($\propto I$ or a Pauli). Fidelity is therefore never graded against the wrong target. Three modes are run: timing-only, phase-only and timing+phase, all initialized at CPMG timing with Y pulse axes in the pre-registered benchmarks. The convergence diagnostic separately examines a different XY-phase timing-only configuration and should not be conflated with that benchmark arm. Each run has a budget of 2000 objective evaluations on 256 frozen training trajectories (common random numbers). There are three independent training runs (training seeds 101–103 with optimizer seeds 1–3).

e) Baselines, selection rule and evaluation: At each n, the baselines are CPMG-n (Y pulses), XY4, XY8 (where n divides), UDD-n and free evolution, with the same T, w, guard, grid, noise model and frame-correction privileges. The *training-selected canonical* baseline is fixed by a pre-registered rule: the lowest *training* infidelity. All arms are then evaluated on the same held-out test draws (seeds 9001–9005, 1024 trajectories each; disjointness from training is enforced in code). We report raw ϵ, the ratio $\epsilon_{\text{selected}}/\epsilon_{\text{CMA}}$ (values > 1 favour CMA-ES), and a two-stage paired bootstrap 95% interval (seeds as blocks, trajectories within seeds). These intervals quantify Monte-Carlo uncertainty within the synthetic model only; they are not significance tests and do not account for model selection or model mismatch.

f) Reproducibility: Each run writes an immutable `results/<run_id>/` directory with a manifest (UTC time, git SHA and dirty flag, platform, Python and package versions, config hash, seeds, command) and SHA-256 checksums. Every table, figure and number macro in this report is regenerated by `scripts/make_report_artifacts.py` from checksum-verified runs pinned in `report/runs.json`; the claim ledger is `report/evidence_map.csv`. The historical full-suite record reports 249 passed, 0 failed, 0 errors, 1 skipped (run `test_suite_20261027T000401Z_2fc21cc36501`; skipped = QUA SDK not installed). Current CI executes the specified smoke suite independently.

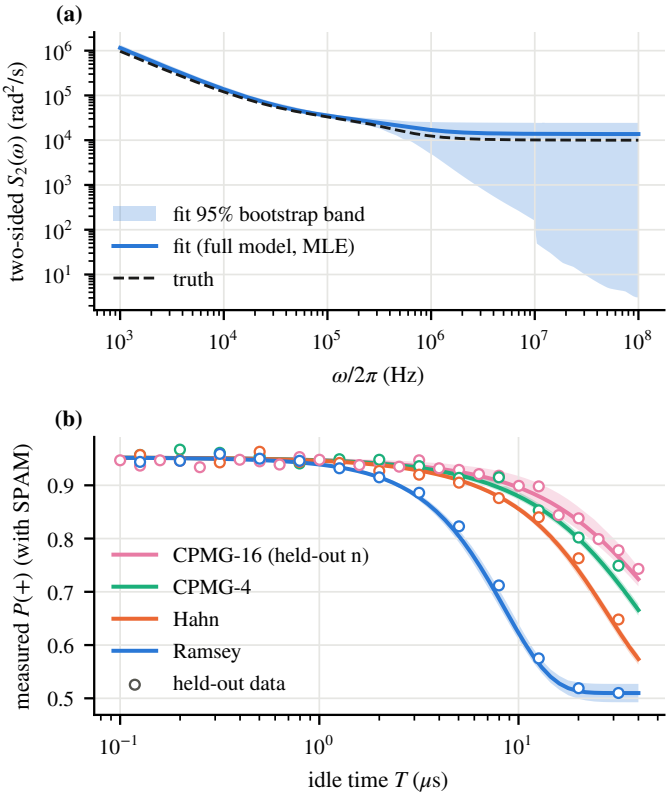


Fig. 1. Synthetic noise spectroscopy. (a) True and fitted two-sided PSD with a 95% bootstrap band; spectral uncertainty is substantial outside the sampled passbands. (b) Fitted decay curves; open circles are independent held-out points only.

V. Experimental Results

All results in this section are computed on *synthetic* device models; no IQCC hardware run has been performed. The experiment configurations, raw outputs, seeds, reproduction commands, checksums and claim ledger are in the public repository (`results/`, `report/evidence_map.csv`). Extended tables, ablation plots and individual trial analyses are included in the supplementary report (`report/sections/results_full.tex`).

A. Simulator and Spectroscopy Validation

In five predetermined configurations, Monte Carlo infidelity agrees with the ideal filter-function reference within the specified combined uncertainty (largest standardized discrepancy $|z| = 0.95$); this establishes internal consistency, *not* device validation. The Gaussian fit predicts 106 held-out synthetic decay measurements (Fig. 1) with RMSE $1.15\times$ the shot-noise floor and 95% interval coverage 0.934. These predictions rely on the same analytic forward family used to generate the data, fixed T_1 and known cutoffs. The spectral model is not uniquely identified.

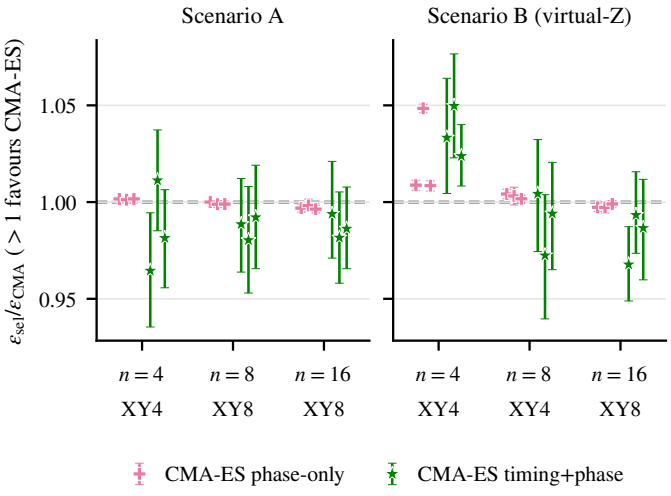


Fig. 2. Held-out ratio $\epsilon_{\mathrm{sel}}/\epsilon_{\mathrm{CMA}}$ (> 1 favours CMA-ES) for each of three CMA-ES training runs, at the same n and $T = 2\,\mu$s as the training-selected baseline named under each n. Bars: paired-bootstrap 95% intervals (Monte-Carlo uncertainty within the model only). Timing-only CMA-ES (ratios 0.071–0.612) is omitted for clarity; raw infidelities are tabulated in the supplement.

B. Scenario A: Canonical DD Versus Optimization

With equal pulse count and independent held-out noise (Fig. 2), no CMA-ES mode provides a practically meaningful improvement over the training-selected XY4/XY8 baseline. The synthetic parameters place canonical DD near a pulse-independent relaxation and white-noise floor, so the available absolute improvement is small even for an ideal optimizer. In particular, the $n = 16$ baseline infidelity 1.00×10^{-2} is close to the simulated T_1-plus-white-noise floor 9.90×10^{-3}. The largest held-out ratio $\epsilon_{\mathrm{sel}}/\epsilon_{\mathrm{CMA}}$ is 1.011 (one $n = 4$ timing+phase run): gains are at most about 1%, which is not practically meaningful and does not establish optimization headroom. Joint timing/phase optimization fits frozen training trajectories more closely while degrading on independent evaluation, consistent with overfitting under the tested budget and model.

C. Scenario B: Readout-Like Detuning

A deterministic, synthetic Stark-like frequency profile was added to the idle window. The experiment does not model stochastic measurement-induced dephasing or measured readout. The joint timing/phase optimization offers a few-percent nominal advantage over XY4 plus virtual-Z at $n = 4$, but this reverses for prescribed ± 100 ns timing shifts. The phase-only $n = 4$ candidate retains a small advantage in the tested perturbation sweep. These positive cells were selected after examining multiple comparisons; they are exploratory and cannot support a general hardware claim.

D. Ablations and Interpretation

The exploratory colored-only case isolates substantially more decouplable noise and is therefore a stronger test of the search strategy. Removing the white-noise and relaxation limits creates additional theoretical headroom, but under the specified search budget CMA-ES gained less than 0.5% over XY8 at $n = 8$ in held-out evaluation (ratios up to 1.004, colored-only phase-only, and 1.004, no-T_1 timing+phase) and lost substantially at $n = 16$ (colored-only timing+phase, ratio 0.394–0.566). The fitted-model pipeline further exposes simulation to model mismatch. Overall, canonical robust DD is a difficult baseline to beat in the tested regimes; the synthetic results do not rule out advantages from alternative noise spectra, optimizers, or real hardware.

VI. Discussion, Limitations and Roadmap

a) Interpretation: The primary pre-registered comparison has limited discriminatory power: under the assumed model, the canonical XY4/XY8 sequences already approach the pulse-independent relaxation and white-noise floor. Thus the small maximum possible improvement should be assessed before searching. Under this pre-registered model the answer to RQ3 is negative. XY4/XY8 already suppress the decouplable colored noise, and the remaining infidelity is set by T_1 and white dephasing, which ideal refocusing pulses do not suppress in this model. Where headroom exists (the colored-only ablation), our CMA-ES setup gained under 0.5% over XY8 at $n = 8$ on held-out noise and lost substantially at $n = 16$. The observed training improvements are consistent with fitting the finite set of frozen trajectories, and the timing+phase search starting from CPMG did not reach XY8's robustness to amplitude errors within 2000 evaluations. These are statements about this parameterization, budget and synthetic model, not about CMA-ES or learned DD in general. The one positive signal (Scenario B, $n = 4$) is a few percent and was identified after the fact. Its timing+phase form does not survive readout-timing uncertainty; its phase-only form does, but only within this model. It would need a new pre-registered test before any claim.

b) Limitations:

- All evidence is synthetic. Device parameters are assumptions, not measurements, and optimizing against a model one also evaluates on is circular; the fitted-model pipeline shows that identification error only adds loss.
- Noise is Gaussian and stationary (except in the telegraph control, which is near-Gaussian at our parameters). Non-stationary, measurement-induced dephasing during readout is *not* modelled; Scenario B is a known deterministic frequency shift only.
- T_1 during pulses is approximated; leakage, crosstalk, finite bandwidth and pulse shaping are absent. Pulse errors are a static per-trajectory amplitude error.
- Monte-Carlo intervals capture sampling uncertainty within the model only; with 5 test seeds the seed-level variability is coarsely resolved. The colored-only, no-T_1, no-amplitude-error and convergence analyses were added after seeing the pre-registered results and are exploratory. One plumbing smoke run touched the test seeds before the main run (documented in `docs/EXPERIMENT_PLAN.md`).

- Robustness of the optimized sequences to calibration drift, Stark rise time and timing-grid quantization (plan item P5.2) was not evaluated. The exploratory ablations reused the held-out test seeds, and the spectroscopy and benchmark bootstraps share one seed.
- Earlier runs recorded the git state at manifest write time rather than run start, and 5 of the 11 pinned runs record a dirty working tree without the uncommitted diff: the pre-registered spectroscopy (G4) run, pre-registered $T = 4\,\mu$s benchmark, no-amplitude-error ablation, no-T_1 ablation, and historical test-suite record. These include pre-registered runs and the source of the spectroscopy headline numbers. Their SHA-256 checksums prove that the stored files are unchanged, not which code produced them. Runs now record the start-of-run git state and diff, which cannot recover the earlier uncommitted code (`docs/REVIEW_FINDINGS.md`).
- The original citation audit environment blocked primary sources; a later audit checked public arXiv reference identities. Publication metadata and final bibliographic presentation are documented in `docs/CITATION_AUDIT.md`.
- No quantum-hardware execution was performed. The QUA adapter is an SDK-free schedule translator marked *not hardware tested*; the QM SDK serialization test was skipped because the SDK is not installed in the project environment.

c) Independent review: An AI red-team pass (`docs/REVIEW_FINDINGS.md`) independently re-derived the conventions, reproduced several reported numbers bit-for-bit and found no critical defect. Its high-severity finding (the Scenario B timing+phase gain is not robust) is incorporated above. Human review of the evidence ledger is pending.

d) Roadmap: (i) Measure, with authorization, T_1, Ramsey, Hahn and CPMG decays and spectator phase during neighbour readout on the target device (`docs/HARDWARE_PROTOCOL.md`), to see whether the decouplable colored component is large enough for optimization to matter at all. (ii) Model measurement-induced dephasing as non-stationary noise and repeat Scenario B. (iii) Regularize the optimizer (more training trajectories, robust objectives, starting from XY8) and pre-register a new held-out benchmark.

Author Contributions

Christian De Zabala F., Madeline Feldner, Zarar Haider, and Udaya Vijay Anand contributed equally to this project, including its research, implementation, evaluation, and report preparation.

Project website: https://udsy19.github.io/quantum-dd-optimizer/

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TABLE S1

GATE G3: MONTE-CARLO (FINITE-PULSE SIMULATOR, 64000 NOISE TRAJECTORIES PER ROW, $dt = 1$ NS) VERSUS FILTER-FUNCTION PREDICTION $r = (1 - e^{-\chi/2})/3$, SCENARIO A GAUSSIAN NOISE WITHOUT T_1 OR CONTROL ERRORS, $T = 2\,\mu$S. PASS CRITERION: $|r_{\mathrm{MC}} - r_{\mathrm{FF}}| \leq 1.96\,\mathrm{SE}$+QUADRATURE ERROR. RUN `g3_validation_20261026T231608Z_cbfe38b59354`.

Configuration	$10^3 r_{\mathrm{FF}}$	$10^3 r_{\mathrm{MC}} \pm \mathrm{SE}$	z	Pass
Free (Ramsey)	21.785	21.673±0.117	−0.95	yes
Hahn, $w = 32$ ns	6.087	6.093±0.033	+0.18	yes
CPMG-8, $w = 32$ ns	3.548	3.553±0.018	+0.27	yes
Hahn, instantaneous	6.081	6.068±0.033	−0.40	yes
CPMG-8, instantaneous	3.518	3.512±0.020	−0.31	yes

SUPPLEMENTARY MATERIAL

The following details and additional tables originate from the same synthetic runs as the main paper. The source artifacts, recorded seeds and checksum manifests are available in the public repository.

APPENDIX A
DETAILED SYNTHETIC EXPERIMENTS AND RESULTS

All results are *simulation predictions for the synthetic model* of Sec. IV; none is a measurement. Each table and figure names the recorded run it comes from; the full claim-to-artifact ledger is `report/evidence_map.csv`.

a) RQ1: Monte Carlo versus filter function (gate G3): Table S1 lists five predetermined configurations: free evolution, Hahn and CPMG-8, the last two with both 32 ns and instantaneous pulses. In every configuration the finite-pulse Monte-Carlo infidelity agrees with the analytic prediction within 1.96 standard errors (64000 trajectories per case; largest $|z| = 0.95$). Halving the noise step changes each estimate by less than the combined statistical error of the two runs. The PSD normalization, filter function, noise sampler and propagator are therefore mutually consistent at the resolution of this test (standard errors of 0.52–0.56% of r). This is a consistency check between two implementations of the same model, not evidence about any device.

b) RQ2: spectroscopy and held-out prediction (gate G4): Figure S1 and Table S2 summarize the pre-registered spectroscopy run. With the correctly specified full Gaussian model, every fitted parameter's 95% bootstrap interval contains the truth. On 106 held-out points (new pulse counts and unseen idle times) the RMSE is 1.15× the shot-noise floor and the 95% predictive coverage is 0.934. The reduced model (no Lorentzian) predicts worse (1.26× floor, coverage 0.877), but our pre-set failure thresholds do not flag it. For this seed the bootstrap intervals for T_2^* and T_2^{Hahn} exclude the true values (9.14 and 28.08 μs), even though each parameter interval covers. With a single recorded seed we cannot tell whether this is chance or a systematic bias of the derived-time intervals; a multi-seed coverage study is future work. This is a favourable test: the data are generated by the same analytic (instantaneous-pulse, Gaussian) forward model that is fitted, T_1 and the $1/f$ cutoffs are fixed at their true values, and only one data seed is recorded. Some parameter intervals are wide (e.g. the white-noise level spans four orders of magnitude), so only the derived decay predictions, not the individual spectral parameters, are well constrained.

The spectrum is *not* uniquely reconstructed. The Lorentzian (σ, τ_c) is strongly correlated and weakly identified; the band above ~ 1 MHz spans orders of magnitude (Fig. S1a); white dephasing and $1/T_1$ are degenerate; and the $1/f$ low cutoff is not identifiable. The telegraph misspecification control is nearly Gaussian at Scenario A parameters (motional narrowing), so it does not stress the Gaussian assumption; the full model fits it well (RMSE 1.05× floor). This control is therefore weak.

c) RQ3: CMA-ES versus canonical DD, Scenario A (pre-registered): Table S3 and Figs. S2 and S3 report held-out infidelities. The training-only rule selects XY4 at $n = 4$ and XY8 at $n = 8, 16$, with $\epsilon = 1.06 \times 10^{-2}$, 1.01×10^{-2} and 1.00×10^{-2}. No CMA-ES mode gives a practically relevant improvement: the best single run (timing+phase, $n = 4$) has ratio 1.011 with an interval that includes 1, and the other modes are within ±0.2% or worse:

- **Phase-only:** ratios 0.996–1.002 (≈ 1). The phases it finds are not XY-type: at $n = 4$ they form a sign-alternated single-axis pattern $(a, \bar{a}, \bar{a}, a)$, and at $n = 8, 16$ they are non-canonical. Several different phase patterns therefore reach the same floor.
- **Timing+phase:** ratios 0.965–1.011 at $n = 4$ and at most 0.994 at $n = 8, 16$.
- **Timing-only:** ratios as low as 0.071. This mode keeps CPMG's Y axes, and CPMG with a 2% amplitude error is far worse than XY4/XY8 under gate-fidelity grading (Table S3).

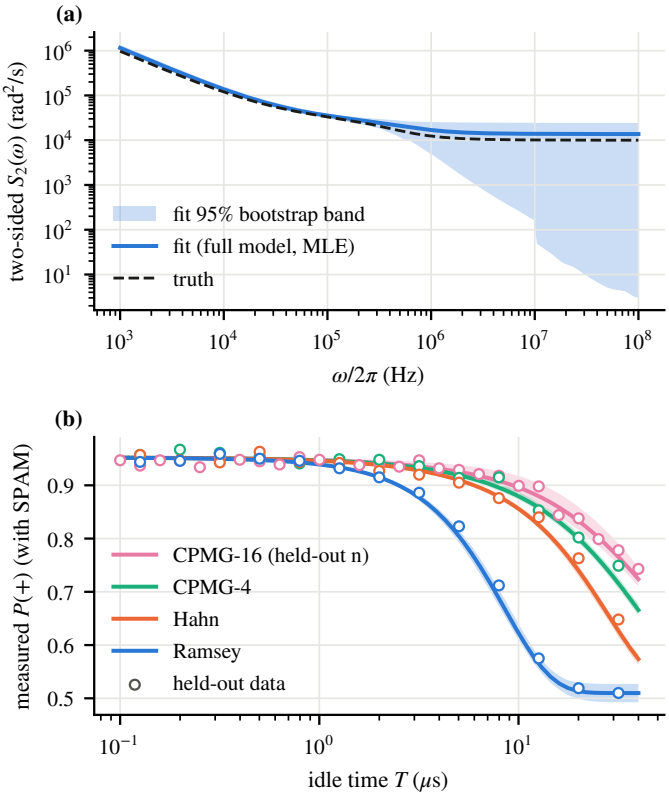


Fig. S1. Synthetic spectroscopy. (a) True versus fitted two-sided PSD (full Gaussian model, MLE) with a 95% parametric-bootstrap band; above ∼1 MHz the data barely constrain the spectrum. (b) Fitted decay curves with 95% bands; open circles are *held-out* data only (CPMG-16 at all times; other families at unseen times). Run as in Table S2.

A major limitation on possible improvement comes from the assumed model. XY8 reaches 1.00×10^{-2}, essentially the pulse-independent floor of 9.90×10^{-3} set by T_1 and the white dephasing component (dashed line in Fig. S2). Under the pre-registered model, no DD sequence can do much better than XY8. At $T = 4\,\mu$s, timing+phase is worse than XY8 (ratio 0.930–0.967 at $n = 16$; Table S5).

d) Overfitting diagnostic: For a single training run at $n = 8$ (exploratory; validation seeds 8001–8002 are disjoint from the test seeds), the training infidelity of timing+phase CMA-ES falls to 9.37×10^{-3}, below XY8's training value. The validation infidelity stays at 1.03×10^{-2}, against 1.03×10^{-2} for XY8 (Fig. S4). The apparent training gain comes from fitting the 256 frozen noise trajectories and does not generalize.

e) RQ4: Scenario B (deterministic Stark-like shift): At $n = 8$, the four pre-registered arms reach $\epsilon = 2.42 \times 10^{-1}$ (no DD, no frame correction), 2.88×10^{-2} (virtual-Z only), 1.02×10^{-2} (best canonical DD + virtual-Z) and 1.01×10^{-2} (CMA-ES + virtual-Z). Most of the benefit comes from frame correction plus any XY-type DD. At the *nominal* Stark profile, CMA-ES beats XY4+virtual-Z only at $n = 4$: phase-only by ratios 1.008–1.048 and timing+phase by 1.024–1.050 across the three training runs. The bracket formed by the three per-run 95% MC intervals is 1.004–1.077 for timing+phase. At $n = 8, 16$ there is no gain (Table S3). These $n = 4$ cases were picked out after inspecting about 27 Scenario B comparisons, so they are exploratory.

The pre-registered robustness sweep (Table S4, Fig. S5) changes the picture for timing+phase. Under ±50–100 ns readout-timing shifts, its worst case

TABLE S2

SYNTHETIC SPECTROSCOPY (GATE G4): HELD-OUT RMSE OF $P(+)$ RELATIVE TO THE SHOT-NOISE FLOOR, 95% PREDICTIVE COVERAGE ON 106 HELD-OUT POINTS, AND $1/e$ TIMES (FIT [95% BOOTSTRAP CI], μS; * = CI EXCLUDES TRUTH). TRUTH: $T_2^* = 9.14\,\mu$S, $T_2^{\mathrm{Hahn}} = 28.08\,\mu$S. "TELEGRAPH" = LORENTZIAN REPLACED BY TELEGRAPH NOISE (MISSPECIFICATION CONTROL). SEEDS 5001/5002/5003, $n_{\mathrm{boot}} = 200$. RUN `spectroscopy_scenario_a_20261026T235246Z_8b272601c344`.

Truth	Fit model	RMSE/floor	Coverage	T_2^*	T_2^{Hahn}	Success
gaussian	full ($\alpha = 1$)	1.15	0.934	8.40 [7.79, 9.08]*	26.48 [24.72, 28.00]*	yes
gaussian	full (α free)	1.15	0.934	8.43 [7.72, 9.04]*	26.14 [24.63, 28.38]	yes
gaussian	reduced	1.26	0.877	8.30 [7.71, 8.89]*	26.52 [25.14, 27.94]*	yes
telegraph	full ($\alpha = 1$)	1.05	0.962	8.69 [8.04, 9.48]	27.12 [25.28, 29.01]	yes
telegraph	reduced	1.16	0.934	8.49 [7.86, 9.09]*	27.29 [25.72, 28.88]	yes

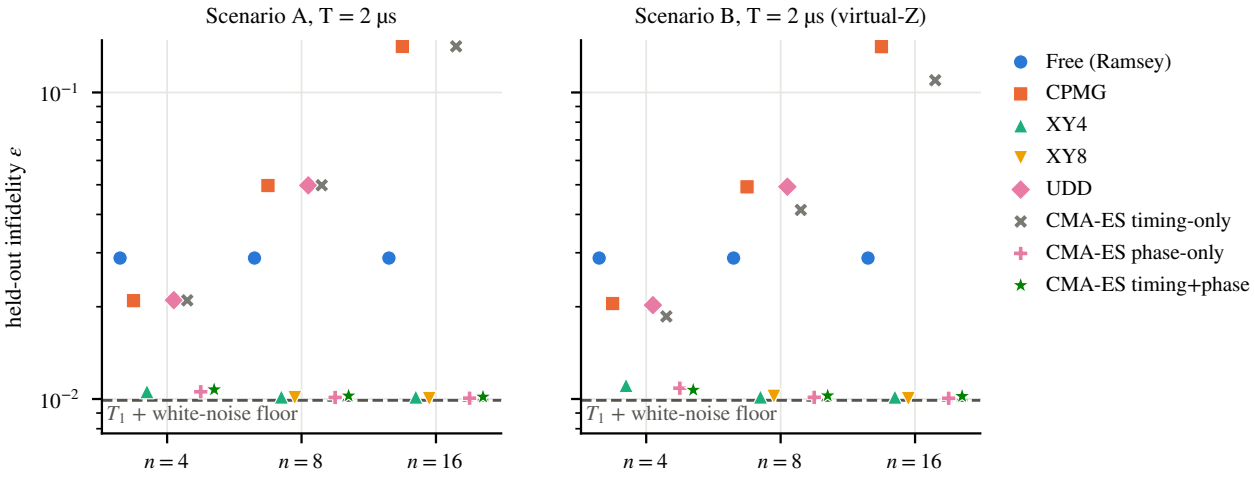


Fig. S2. Held-out infidelity at equal n, $T = 2\,\mu$s, $w = 32$ ns (mean over test seeds 9001–9005, 1024 trajectories each; CMA-ES: mean over 3 training runs; MC standard errors are smaller than the markers). The dashed line is the analytic floor from T_1 and white noise. Scenario B arms use virtual-Z from the nominal Stark profile. Near the floor the markers overlap; Fig. S3 shows the comparison with the selected baseline as a ratio.

TABLE S3

HELD-OUT BENCHMARK AT $T = 2\,\mu$S, $w = 32$ NS: INFIDELITY $10^3\epsilon$ AVERAGED OVER TEST SEEDS 9001–9005 (1024 TRAJECTORIES EACH; MC STANDARD ERROR OF EACH 5-SEED MEAN $\leq 1.5\%$ OF ϵ, PER SEED $\leq 3.5\%$). "SEL." = BASELINE SELECTED BY THE PRE-REGISTERED TRAINING-ONLY RULE. CMA-ES: MEAN ϵ OVER 3 INDEPENDENT TRAINING RUNS AND, IN PARENTHESES, THE RANGE OF $\epsilon_{\mathrm{sel}}/\epsilon_{\mathrm{CMA}}$ OVER RUNS (> 1 FAVOURS CMA-ES). SCENARIO B ARMS ALL USE VIRTUAL-Z FROM THE NOMINAL STARK PROFILE. UDD-16 IS INFEASIBLE ON THE GRID (TABLE S7). RUNS: `benchmark_scenario_a_20261026T232517Z_1bd28ade742a`; `benchmark_scenario_b_20261026T234545Z_ad7b16de1c23`.

		Canonical baselines						CMA-ES (ratio range)		
Scenario	n	Free	CPMG	XY4	XY8	UDD	Sel.	timing-only	phase-only	timing+phase
A	4	28.8	20.9	10.6	—	21.0	XY4	21.0 (0.500–0.507)	10.6 (1.001–1.002)	10.7 (0.965–1.011)
	8	28.8	49.7	10.2	10.1	49.8	XY8	49.8 (0.203–0.203)	10.1 (0.999–1.000)	10.2 (0.980–0.992)
	16	28.8	141.3	10.1	10.0	—	XY8	141.4 (0.071–0.071)	10.1 (0.996–0.998)	10.2 (0.982–0.994)
B (virtual-Z)	4	28.8	20.5	11.1	—	20.2	XY4	18.6 (0.568–0.612)	10.8 (1.008–1.048)	10.7 (1.024–1.050)
	8	28.8	49.2	10.2	10.2	49.3	XY4	41.4 (0.245–0.246)	10.1 (1.002–1.004)	10.3 (0.972–1.004)
	16	28.8	141.1	10.1	10.0	—	XY8	109.6 (0.090–0.093)	10.1 (0.997–0.999)	10.2 (0.968–0.993)

reaches 2.11×10^{-2}, against 1.13×10^{-2} for XY4+virtual-Z, and it wins in only 3–8 of the 10 perturbed profiles. At $n = 8$, two of three runs lose in all 10. This nominal gain therefore does *not* survive the uncertainty RQ4 asks about. Only the $n = 4$ phase-only sequences remain marginally better than XY4+virtual-Z in all 10 perturbations in all three runs (worst case 1.12×10^{-2} versus 1.13×10^{-2}), a few-percent, model-specific effect. Virtual-Z alone degrades to 3.86×10^{-2} under a $\pm 20\%$ amplitude error. Timing shifts leave it unaffected, because the integrated phase of a shift that stays inside the window is unchanged.

f) RQ5: robustness, ablations and model mismatch: Table S5 adds post-hoc ablations, labelled exploratory. Removing T_1 and the white floor ("colored only") opens two orders of magnitude of headroom, since XY8 then reaches 9.75×10^{-5} at $n = 16$. Even so, timing+phase CMA-ES under the same 2000-evaluation budget loses to XY8 (ratio 0.394–0.566), and phase-only gains at most 1.010 at $n = 4$. Within this budget and parameterization (starting from CPMG, with $2n$ free coordinates), CMA-ES did not find sequences that generalize better than XY-family designs. The training-only selection rule can itself pick a baseline that is slightly worse on test (UDD in the no-amplitude-error ablation), which inflates ratios there by $< 1\%$. Robustness to calibration drift, Stark rise time and timing-grid quantization of the optimized sequences (master-plan item P5.2) was *not* run. In the model-based pipeline (Table S6), optimizing on the noise model *fitted* from synthetic spectroscopy and testing on the truth gives ratios 0.982–1.011 (full model, $n = 4$) and 0.894–0.984 (reduced model): model error adds further loss, and with the reduced model the baseline-selection rule sometimes picks the weaker family.

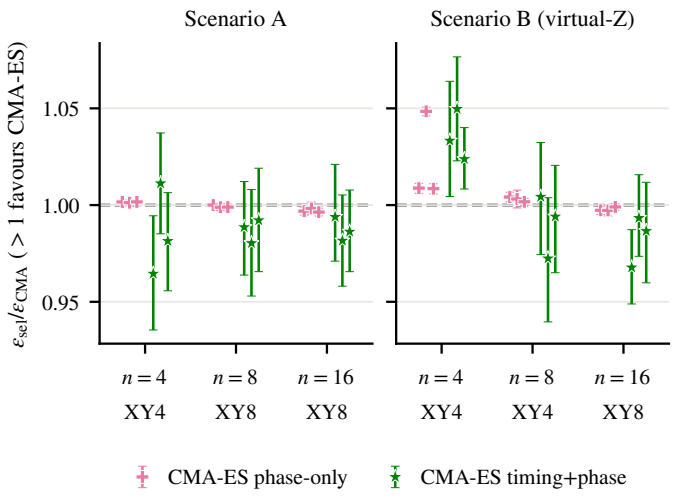


Fig. S3. Held-out ratio $\epsilon_{sel}/\epsilon_{CMA}$ (> 1 favours CMA-ES) for each of the three training runs, with pooled paired-bootstrap 95% intervals; the training-selected baseline is named under each n. Timing-only CMA-ES (ratios 0.071–0.612) is omitted for clarity; see Table S3.

TABLE S4

SCENARIO B ROBUSTNESS (PRE-REGISTERED SWEEP): WORST-CASE $10^3\epsilon$ OVER 10 PERTURBED STARK PROFILES (AMPLITUDE $\times\{0.8, \ldots, 1.2\}$, TIMING $\pm\{50, 100\}$ NS AND NOMINAL), VIRTUAL-Z ALWAYS FROM THE NOMINAL PROFILE; CMA-ES VALUES FOR TRAINING RUNS 101/102/103, AND IN HOW MANY OF THE 10 PERTURBATIONS CMA-ES BEATS THE SELECTED DD + VIRTUAL-Z. RUN `benchmark_scenario_b_20261026T234545Z_ad7b16de1c23`.

n	CMA-ES mode	VZ only	DD+VZ	CMA-ES+VZ (runs)	wins/10
4	phase-only	38.6	11.3 (XY4)	11.2 / 10.6 / 11.2	10 / 10 / 10
4	timing+phase	38.6	11.3 (XY4)	16.0 / 14.5 / 21.1	8 / 6 / 3
8	phase-only	38.6	10.2 (XY4)	10.4 / 10.5 / 10.4	8 / 8 / 7
8	timing+phase	38.6	10.2 (XY4)	10.7 / 19.9 / 20.8	9 / 0 / 0
16	phase-only	38.6	10.0 (XY8)	10.4 / 10.1 / 10.2	0 / 0 / 0
16	timing+phase	38.6	10.0 (XY8)	15.0 / 11.2 / 10.5	0 / 0 / 0

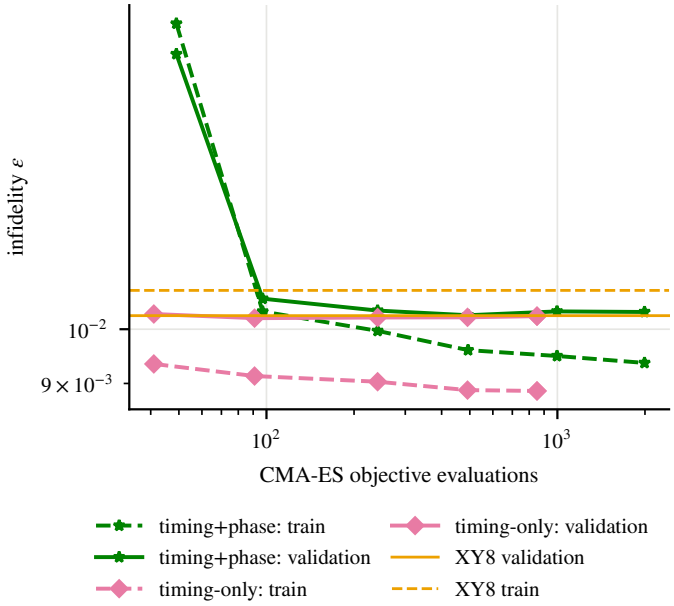


Fig. S4. Training (dashed) versus validation (solid) infidelity of the best-so-far CMA-ES candidate as a function of evaluation budget; Scenario A, $n = 8$, training seed 101. Timing+phase starts from CPMG. The timing-only curve in this exploratory plot was labeled with XY phases, whereas the pre-registered timing-only arm fixes CPMG-Y phases. Until the convergence-plot configuration is independently reconciled, this curve must not be used as evidence for the pre-registered timing-only benchmark. The horizontal lines show XY8. Exploratory diagnostic.

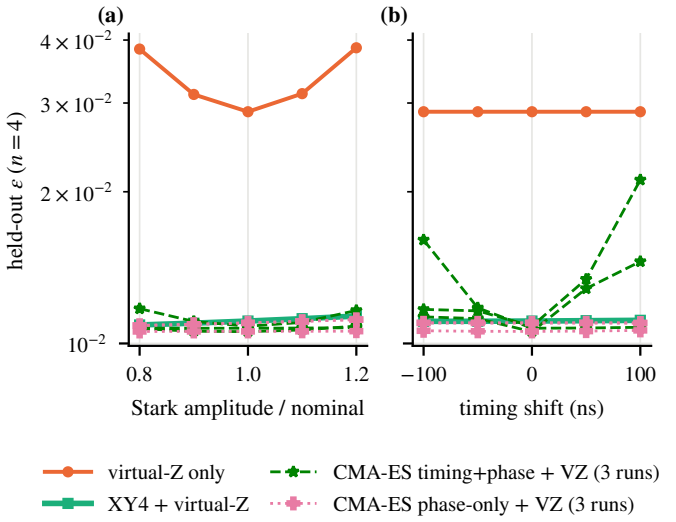


Fig. S5. Scenario B robustness at $n = 4$, all three training runs: (a) Stark amplitude and (b) readout timing shift perturbed in evaluation while every virtual-Z angle uses the nominal profile. Timing+phase CMA-ES (dashed) degrades under timing shifts; phase-only (dotted) does not.

TABLE S5

HELD-OUT COMPARISON ACROSS SCENARIO VARIANTS (TEST SEEDS 9001–9005). RATIO $= \epsilon_{\text{selected}}/\epsilon_{\text{CMA}}$ (RANGE OVER 3 TRAINING RUNS; > 1 FAVOURS CMA-ES). FLOOR = ANALYTIC T_1 AMPLITUDE DAMPING COMPOSED WITH WHITE DEPHASING OVER T (PULSE-INDEPENDENT). †POST-HOC EXPLORATORY ABLATIONS ADDED AFTER SEEING THE PRE-REGISTERED RESULTS.

Variant	n	Sel.	$10^3\epsilon_{\text{sel}}$	$10^3\times$floor	timing-only	phase-only	timing+phase
A (pre-registered), $T = 2\,\mu$s	4	XY4	10.6	9.90	0.500–0.507	1.001–1.002	0.965–1.011
	8	XY8	10.1	9.90	0.203–0.203	0.999–1.000	0.980–0.992
	16	XY8	10.0	9.90	0.071–0.071	0.996–0.998	0.982–0.994
A, $T = 4\,\mu$s (pre-registered)	4	XY4	22.8	19.6	0.693–0.699	1.000–1.000	0.975–1.003
	8	XY8	20.6	19.6	0.349–0.352	0.999–1.000	0.939–0.981
	16	XY8	19.9	19.6	0.136–0.136	0.997–0.998	0.930–0.967
A without amplitude error†	4	CPMG/UDD	10.6/10.6	9.90	0.969–1.007	0.999–1.008	0.969–1.018
	8	UDD	10.2	9.90	0.993–1.007	1.007–1.009	0.992–1.006
	16	CPMG/XY8	10.0/10.0	9.90	0.987–0.998	0.998–0.999	0.983–0.997
A without T_1 †	4	XY4	4.00	3.32	0.271–0.277	1.003–1.004	0.917–1.031
	8	XY8	3.53	3.32	0.080–0.081	0.996–1.001	0.953–1.004
	16	XY8	3.45	3.32	0.025–0.025	0.991–0.995	0.937–0.994
A, colored noise only†	4	XY4	0.631	0.000	0.056–0.057	1.006–1.010	0.967–0.986
	8	XY8	0.238	0.000	0.006–0.006	1.002–1.004	0.894–0.960
	16	XY8	0.098	0.000	0.001–0.001	0.901–0.954	0.394–0.566

TABLE S6

MODEL-BASED PIPELINE: BASELINE SELECTION AND CMA-ES (TIMING+PHASE) TRAINED ON THE NOISE MODEL IN COLUMN 1, ALL ARMS EVALUATED ON THE *true* SCENARIO A MODEL (TEST SEEDS 9001–9005). FITTED MODELS COME FROM RUN `spectroscopy_scenario_a_20261026T235246Z_8b272601c344`. RATIO AS IN TABLE S5; CI = ENVELOPE OF POOLED PAIRED-BOOTSTRAP 95% INTERVALS OVER THE 3 TRAINING RUNS. RUN `model_based_scenario_a_20261026T235628Z_373dfea37a30`.

Training model	n	Sel.	$10^3\epsilon_{\text{sel}}$	$10^3\epsilon_{\text{CMA}}$	Ratio	CI envelope
truth	4	XY4	10.6	10.7	0.965–1.011	[0.936, 1.037]
fitted full	4	XY4	10.6	10.6	0.982–1.011	[0.960, 1.036]
fitted reduced	4	XY4	10.6	11.1	0.894–0.984	[0.873, 1.005]
truth	8	XY8	10.1	10.2	0.980–0.992	[0.952, 1.019]
fitted full	8	XY8	10.1	10.3	0.966–0.991	[0.927, 1.016]
fitted reduced	8	XY4/XY8	10.1	10.7	0.896–0.986	[0.868, 1.011]

TABLE S7

TIMING FEASIBILITY, SQUARE π PULSES $w = 32$ NS, GUARD 4 NS, PULSE STARTS SNAPPED TO A 4 NS GRID (ASSUMED CONTROLLER TICK). "SHIFT" = LARGEST CHANGE OF A PULSE CENTRE CAUSED BY GRID SNAPPING; "MIN GAP" = SMALLEST FREE GAP OVER FEASIBLE FAMILIES. INFEASIBLE: EDGE PULSES VIOLATE THE GUARD AFTER SNAPPING. CONTROL CHECK: 16×40 NS PULSES IN A 500 NS WINDOW IS REFUSED BY VALIDATION.

T (μs)	n	Feasible	Infeasible	Max shift (ns)	Min gap (ns)
2	4	CPMG, XY4, UDD	—	2.0	176
2	8	CPMG, XY4, XY8, UDD	—	2.0	44
2	16	CPMG, XY4, XY8	UDD	1.5	48
4	4	CPMG, XY4, UDD	—	2.0	364
4	8	CPMG, XY4, XY8, UDD	—	2.0	104
4	16	CPMG, XY4, XY8, UDD	—	2.0	20